OpenStudio / Studio13-v3 Feature Inventory

> **Last updated:** 2026-06-01
>
> **Primary project format:** `.osproj`
>
> **Legacy project format:** `.s13`
>
> **Scope of this update:** current source tree, mounted UI, action registry, frontend bridge, backend native functions, and adjacent feature docs.
>
> **Note:** This file used to be a historical REAPER-parity tracker with stale counts. It is now a current feature inventory. Keep older planning docs for traceability, but use this file, `README.md`, and `docs/implemented_features.md` for public-facing feature claims.

Status Legend

Status	Meaning
Implemented	User-facing workflow has a UI/action and a store, bridge, or backend path.
Partial / Experimental	Real code exists, but the workflow is incomplete, hardware/runtime dependent, or not yet release-hardened.
Planned / Stub	UI state, bridge stubs, or planning notes exist, but the feature should not be advertised as complete.

Latest Additions Since The Old Tracker

Area	Latest feature surface
AI runtime setup	In-app AI Tools setup with install, cancel, reset, status refresh, feature selection, background install notification, and platform-specific runtime plans.
AI music generation	ACE-Step 1.5 XL Turbo workflows for text-to-music and lyrics-plus-style generation.
AI audio generation	Stable Audio 3 Medium support for text-to-audio and source-conditioned workflows, including license-gated local model import.
Clip AI workflows	Create variation, inpaint selected range, and continue clip from a selected source audio clip.
Stem separation	Async stem separation with progress polling, cancellation, selectable stems, and result import to tracks.
Audio-to-MIDI	Convert an audio clip to a new MIDI track using polyphonic note extraction where the native Basic Pitch / ONNX path is available.
Detached editors	Detached mixer and detached MIDI editor window flows with UI snapshot publishing through the native bridge.
MIDI editor depth	Docked or windowed piano roll sessions, velocity/CC lanes, pitch bend controls, quantize, note transforms, and range editing.
Render/export	Dither path, secondary output, render queue, region render matrix, DDP export, batch converter, render-in-place, and add-render-to-project options.
Routing and mixing	Send/bus routing, routing matrix, sidechain assignment, channel output selection, stereo width, phase invert, pan law, and mixer snapshots.
Project delivery	Project compare, archive/unarchive, clean project directory, safe-mode open, project templates, and project tabs.
Video and sync	Video window plumbing, FFmpeg-backed video/audio extraction path, MTC input/output, MIDI clock sync state, and timecode settings.
Scripting and effects	Lua script editor/API, S13FX/JSFX audio effects, JSFX `@gfx` editor support, plugin A/B states, FX chain presets, and built-in FX presets.
Workflow customization	Command palette, shortcuts modal, screensets, theme editor, toolbar editor, custom actions/macros, mouse modifiers, and help surfaces.

At A Glance

Area	Current support
Core engine	JUCE C++ audio engine with React/TypeScript WebView UI and synchronous `window.__JUCE__.backend` bridge.
Recording	Multitrack audio recording, MIDI recording, punch range, record modes, armed tracks, record-safe state, and input monitoring.
Editing	Clip move/trim/split, ripple edit, razor edit, time selection edits, fades, grouping, takes, slip edit, reverse, normalize, time stretch, and pitch shift.
MIDI	MIDI and instrument tracks, piano roll, virtual keyboard, MIDI import/export, MIDI output routing, quantize, transforms, CC lanes, and audio-to-MIDI.
Mixing	Mixer, detached mixer, sends, buses, routing matrix, master strip, channel strip EQ, metering, automation, mixer snapshots, and sidechains.
Plugins	VST3 hosting, CLAP/LV2 code paths, native editors, input/track/master/monitoring FX, presets, A/B states, MIDI learn, and built-in FX.
Pitch	Graphical pitch editor, YIN pitch analysis, note blobs, drift/vibrato tools, real-time pitch corrector, polyphonic detection, and ARA host plumbing.
AI	Optional local AI Tools runtime for stem separation, ACE-Step music generation, Stable Audio 3 audio generation, variation, inpaint, and continuation.
Render	WAV, AIFF, FLAC, MP3, OGG, stems, selected items, razor areas, regions, render queue, region matrix, DDP, batch conversion, dither, metadata, and secondary output.
Project tools	`.osproj` save/load, `.s13` legacy support, recent projects, templates, safe mode, autosave/backup, compare, archive, clean directory, missing media resolver.
Workflow	Command palette, keyboard shortcuts, menus, screensets, themes, toolbar editor, help overlay, getting started guide, big clock, and timecode settings.
Sync/media	MTC/MIDI clock plumbing, control surface manager, OSC support, MCU-style control surface paths, video window, and FFmpeg video extraction.

Core Engine, Transport, And Recording

Implemented

- Native JUCE audio engine with a React/TypeScript UI hosted in WebView2.
- Audio device settings for driver, device, input/output channels, sample rate, buffer size, and channel configuration.
- Sample-rate-aware clip playback and render mixing through the playback engine.
- Transport controls for play, pause, stop, record, seek, rewind, loop, set loop to time selection, and auto-scroll.
- Tempo, tap tempo, time signature, tempo markers, and metronome controls.
- Metronome accenting, volume, custom click/accent sounds, reset sounds, and render-to-file support.
- Multitrack audio recording with track arm, record-safe, input channel selection, monitoring, punch range, and record modes.
- MIDI recording with live preview and completed MIDI clip handoff.
- Background waveform peak cache and recording waveform previews.

- Real-time meter update event flow isolated from track arrays for render performance.

Partial / Experimental

- External sync paths exist for MTC and MIDI clock, but device and studio integration still needs hardware validation.
- LTC output has a UI/bridge surface, but the backend explicitly treats generation as a stub.
- Live capture output has bridge/store state and should be treated as experimental until fully validated in-session.

Project, Files, And Media Management

Implemented

- New, open, save, save as, close project, quit, and unsaved-changes confirmation flows.
- `.osproj` as the primary project extension with `.s13` legacy open/save support.
- Recent projects, startup recovery/diagnostics, and safe-mode project open with FX bypass.
- Project settings for name, notes, sample rate, bit depth, tempo, time signature, author/revision style metadata, and related state.
- Timestamped backup/autosave preferences.
- Project tabs, project templates, save-as-template, and new-from-template flows.
- Project compare with saved version.
- Session archive/unarchive backend paths.
- Media import through native dialogs, drag/drop, and audio/video import path.
- Missing media resolver.
- Media explorer surface, recent paths, and media import workflow.
- Clean project directory tool.
- Batch file converter modal.

Partial / Experimental

- Media explorer audition/preview should be treated as limited until the backend preview engine is expanded.
- AAF/session interchange code paths exist, but AAF import is not a complete advertised workflow.

Arrangement And Clip Editing

Implemented

- Konva timeline with ruler, grid, snap, zoom, scroll, playhead, waveform rendering, MIDI thumbnails, and time selection.
- Audio and MIDI clip creation, media import, drag/drop, move, resize, trim, and track-to-track moves.
- Multi-clip and multi-track selection, select all clips, select all tracks, and deselect all.
- Undo/redo through the command manager for editing operations.
- Cut, copy, paste, duplicate, delete, nudge, and fine nudge.
- Split at cursor and split at time selection.
- Cut/copy/delete within time selection and insert silence.
- Ripple editing modes: off, per-track, and all-tracks.
- Razor edit areas with content deletion and render source support.
- Clip mute, lock, color, volume, pan, fade in/out, fade shape, gain envelope, and clip properties panel.
- Auto-crossfade and a crossfade editor surface.
- Reverse clip, normalize selected clips, time stretch, and clip pitch shift.
- Dynamic split/transient split UI and execution path.

- Slip editing and free item positioning.
- Takes: add take, set active take, explode takes to tracks, and implode clips into takes.
- Track spacers, empty items, empty MIDI items, insert multiple tracks, and folder tracks.
- Clip launcher view.
- Quantize selected clips to grid.

MIDI And Instruments

Implemented

- MIDI device enumeration, input opening, output routing, and MIDI panic.
- MIDI tracks and instrument tracks.
- Virtual instrument on new track and quick-add instrument track actions.
- MIDI clip storage, serialization, playback scheduling, import, export, and project MIDI export.
- Piano roll editor with docked and detached/windowed sessions.
- Note draw, edit, select, cut/copy/paste, delete, resize, move, and range operations.
- Velocity editing, CC lanes, pitch bend controls, visible lane preferences, and note inspector style surfaces.
- MIDI quantize using last settings, reset quantize, freeze quantize, and quantize test coverage.
- MIDI transforms: transpose, octave transpose, velocity scale, reverse notes, invert note pitches, and snap selected notes to scale.
- Virtual piano keyboard.
- MIDI input readiness checks before recording.
- Audio-to-MIDI conversion creates a generated MIDI track beside the source track, with undo support.
- Built-in instrument paths for basic synth, piano, drums, and sampler-style state.

Partial / Experimental

- Drum editor and media pool toggles/actions exist, but should be confirmed as mounted, complete workflows before being advertised heavily.
- Step sequencer/step input state exists in the store surface, but the full user workflow needs validation.

Mixing, Routing, Metering, And Automation

Implemented

- Mixer panel, channel strips, master strip, master track in TCP, and detached mixer window.
- Track controls for volume, pan, mute, solo, arm, input monitoring, record-safe, channel count, playback offset, and output channels.
- Master volume, pan, mute, mono, and master automation state.
- Peak/RMS meters, master meter cluster, clipping state, and reset meter clip actions.
- Channel strip EQ modal and built-in channel EQ backend parameters.
- Sends with level, pan, enable, pre/post fader, and phase invert.
- Bus/group tracks and create-bus-from-selected-tracks.
- Routing matrix and track routing modal.
- Sidechain source assignment into plugins.
- Phase invert, stereo width, master send enable, pan law, and DC offset handling.
- Track groups / linked group parameters.
- Mixer snapshots with save, recall, delete, backend sync, and undo on recall.

- Track and master automation lanes with read, write, touch, latch, manual point editing, range replace, backend sync, and envelope manager.
- Move-envelopes-with-items option.
- Loudness meter, phase correlation, and spectrum data bridge paths.

Plugins, FX, And Scripting

Implemented

- Plugin scanning and loading for VST3, with CLAP/LV2 code paths present.
- Native plugin editor window management.
- Input FX, track FX, master FX, and monitoring FX chains.
- Add, remove, bypass, reorder, and open editor flows for FX chains.
- Plugin parameters, preset load/save, state save/load, and plugin A/B compare.
- Plugin MIDI learn, mapping list, and clear mapping flows.
- FX chain presets for track, input, and master chains.
- Built-in EQ, compressor, gate, limiter, delay, reverb, chorus, saturator, pitch corrector, and instrument-style processors.
- Built-in FX preset save/load/delete.
- Built-in FX oversampling controls.
- S13FX / JSFX-style script effects.
- JSFX `@gfx` native editor support through `S13FXGfxEditor`.
- Lua script execution, script directory/listing, script editor UI, and console output.
- App-facing Lua API documentation in `docs/API.md`.

Partial / Experimental

- CLAP and LV2 support should be described as code-path/plugin-format support until compatibility is validated with a plugin test matrix.
- 32-bit plugin bridge is currently a preference/toggle surface, not a completed out-of-process legacy plugin host.

Pitch, Analysis, And Audio Processing

Implemented

- Monophonic pitch analysis with YIN contour and note segmentation.
- Graphical pitch editor with note blobs, pitch contour, piano grid, zoom, scroll, and lower-zone UI.
- Pitch editor tools for pitch, drift, vibrato, transition, draw/split style editing, selection, undo, and redo.
- Scale/key snapping, chromatic snap, correct-pitch macro, and scale detection support.
- Offline monophonic graphical pitch correction/apply path.
- Pitch preview, scrub preview, rendered preview segments, and route/status diagnostics.
- Real-time auto-tune style pitch corrector FX.
- Polyphonic detection and audio-to-MIDI extraction through the Basic Pitch / ONNX path where available.
- Audio analyzer, transient detection, and silent-region style analysis utilities.
- ARA host controller lifecycle and track/clip ARA status plumbing.

Partial / Experimental

- Polyphonic pitch correction/resynthesis is less mature than monophonic pitch editing. Treat detection and MIDI extraction as stronger than polyphonic audio correction.

- Subjective pitch, formant, stem, and artifact quality always requires user audition. Harness diagnostics are not proof of audible quality.

AI And Assisted Audio

Implemented

- Optional AI Tools runtime status with refresh, install, cancel, reset, selected feature install, requested feature routing, and setup progress.
- Modular AI Tools feature IDs for `stemSeparation` and `audioGeneration`.
- Hardware/runtime status including GPU support hints and model/runtime readiness.
- Platform runtime packaging plans for Windows DirectML/CUDA and Linux CUDA/ROCm style profiles.
- Stem separation workflow with selectable stems, progress polling, cancellation, and imported result clips/tracks.
- AI track type and AI track header controls.
- AI generation modal and AI workflow parameter fields.
- ACE-Step 1.5 XL Turbo model surface.
- Stable Audio 3 Medium model surface.
- AI workflows:
- Text to Music
- Lyrics + Style
- Text to Audio
- Create Variation
- Inpaint Selection
- Continue Clip
- Source-conditioned AI generation from selected clips, with time-selection requirements for inpaint.
- AI generation progress and cancellation.
- Stable Audio setup includes local snapshot selection and license acknowledgement state.

Partial / Experimental

- AI generation quality and speed depend on installed optional runtimes, local hardware, available VRAM/RAM, model availability, and accepted model licenses.
- The core app intentionally does not bundle large AI model runtimes.

Render, Export, And Delivery

Implemented

- Offline render through the same playback/FX path used by playback.
- Render sources for master, stems, selected tracks, selected media items, selected items through master, and razor edit areas.
- Region render matrix UI.
- Render bounds for entire project, custom range, time selection, project regions, and selected regions where the UI path is available.
- Output directory, filename, and wildcard filename resolution.
- Metadata state for rendered files.
- Formats: WAV, AIFF, FLAC, MP3, and OGG.
- Sample-rate conversion and lossy encoding through FFmpeg where needed.

- Mono/stereo output, bit depth/quality options, normalize, render tail, and include-metronome options.
- Dithered render path through `renderProjectWithDither`.
- Secondary output format and secondary bit depth.
- Online render and add-rendered-output-to-project UI state.
- Render queue panel and queue actions.
- Render clip in place and render track in place.
- Consolidate track.
- Freeze/unfreeze track state and related UI.
- DDP disc image export backend and modal.
- Batch converter modal.
- Export project MIDI.

Partial / Experimental

- Region render matrix, DDP, batch conversion, online render, and add-to-project workflows should receive release smoke testing before high-confidence public release notes.
- Advanced broadcast metadata, immersive multichannel delivery, and full CD mastering validation remain specialist areas to test separately.

Workflow, UI, And Customization

Implemented

- Central action registry used by menus, command palette, shortcut reference, and global shortcuts.
- Menu bar, main toolbar, transport bar, lower zone, mixer, timeline, and modal surfaces.
- Command palette.
- Keyboard shortcuts modal.
- Preferences modal with general, editing, display, and backup preferences.
- Help overlay and getting started guide.
- Big clock and timecode settings.
- Screensets/layout state with save/load actions.
- Theme presets, theme editor, custom theme overrides, theme import/export style state, and high-contrast support.
- Toolbar editor and custom toolbar state.
- Custom actions/macros.
- Mouse modifier preferences and reset.
- Toast notifications, modal safety guards, startup recovery app, and error boundary.
- Detachable panels for mixer and MIDI editor.

Sync, Control Surfaces, Video, And Pro-Audio Plumbing

Implemented / Present

- Timecode display and timecode settings panel.
- MTC generator and receiver paths.
- MIDI clock sync source/status plumbing.
- MIDI control surface manager.
- OSC control surface support.
- MCU-style control surface support.

- Video window component.
- FFmpeg-backed video metadata/frame extraction and audio extraction path.
- Surround/channel layout and VBAP panner source modules.
- Track channel format and master channel format state.

Partial / Experimental

- Video support is functional plumbing, not a full post-production video suite.
- External sync and control surface workflows should be tested with real devices.
- Surround/immersive workflows need dedicated validation before being advertised as mature.

File Types And Sidecar Assets

Type	Current role
`.osproj`	Primary project file.
`.s13`	Legacy project file support.
`.ostheme`	Current theme export/import target.
`.s13theme`	Legacy theme import support.
`.ospreset`	Built-in FX preset style.
`.ospeaks`	Current waveform peak cache sidecar.
`.s13peaks`	Legacy waveform peak cache sidecar support.
`.mid` / `.midi`	MIDI import/export.
`.wav`, `.aiff`, `.flac`, `.mp3`, `.ogg`	Render/export and media workflows.

Current Caveats To Keep Honest

Feature	Caveat
Polyphonic pitch correction	Detection and MIDI extraction are present; polyphonic audio correction/resynthesis is still less mature.
CLAP/LV2 hosting	Code paths exist; plugin compatibility still needs broad validation.
32-bit plugin bridge	Toggle/state exists; full legacy plugin bridge is not complete.
LTC output	Bridge surface exists; backend generation is stubbed.
Media explorer preview	Import/browse surface exists; audition preview is limited.
AAF import/interchange	Source code exists, but it should not be advertised as complete AAF support.
Video	FFmpeg-backed video plumbing exists; editing/post-production workflow maturity is still evolving.
AI generation	Optional runtime and hardware dependent; model licenses and local setup affect availability.
Audio quality claims	Pitch, formant, stem, and generation quality must be validated by listening, not just diagnostics.

Source Cross-Reference

- `README.md` - public overview and product positioning.
- `docs/implemented_features.md` - codebase feature audit and caveats.
- `frontend/src/store/actionRegistry.ts` - commands exposed to menus, shortcuts, and command palette.
- `frontend/src/App.tsx` - mounted panels, modals, lower zones, detached windows, and workflow surfaces.
- `frontend/src/services/NativeBridge.ts` - frontend-to-native feature boundary.
- `frontend/src/data/aiWorkflows.ts` - AI models, workflows, and parameters.
- `frontend/src/store/actions/*` - current Zustand action modules.
- `Source/MainComponent.cpp` - native bridge functions exposed to the frontend.
- `Source/AudioEngine.*`, `Source/PlaybackEngine.*`, `Source/TrackProcessor.*`, `Source/MIDIManager.*`, `Source/Pitch*`, `Source/StemSeparator.*`, `Source/AITrackEngine.*`, and related source files - backend feature implementation.